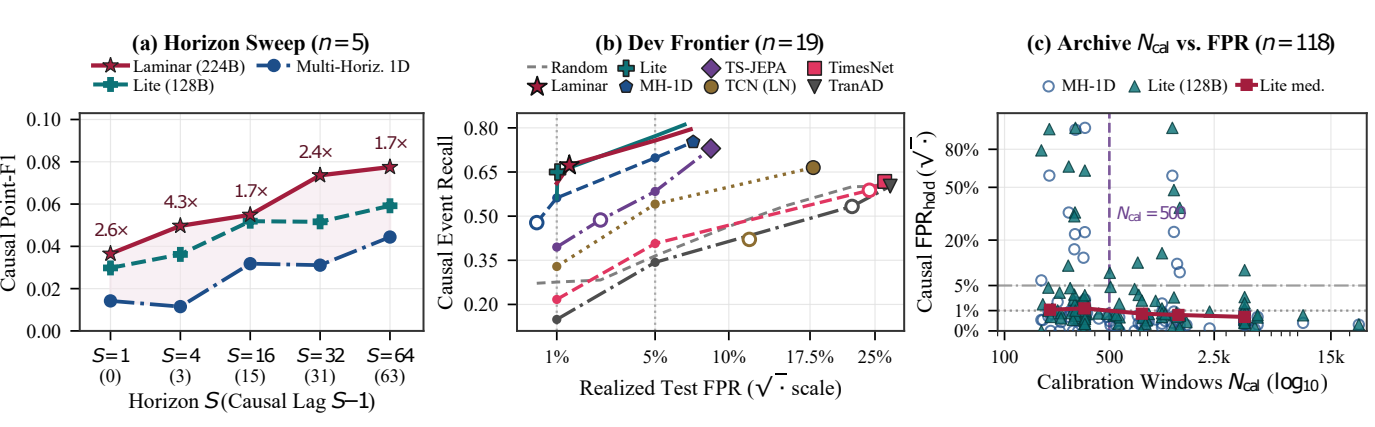

(a) Horizon Sweep ($n=5$)
Laminar (224B)
Multi-Horiz. 1D
Lite (128B)
Causal Point-F1
0.10
0.08
0.06
0.04
0.02
0.00
2.6×
4.3×
1.7×
2.4×
1.7×
$S=1$ (0)
$S=4$ (3)
$S=16$ (15)
$S=32$ (31)
$S=64$ (63)
Horizon S (Causal Lag $S-1$)
(b) Dev Frontier ($n=19$)
Random
Lite
TS-JEPA
TimesNet
Laminar
MH-1D
TCN (LN)
TranAD
Causal Event Recall
0.80
0.65
0.50
0.35
0.20
1%
5%
10%
17.5%
25%
Realized Test FPR ($\sqrt{\cdot}$ scale)
(c) Archive N_{cal} vs. FPR ($n=118$)
MH-1D
Lite (128B)
Lite med.
Causal $\mathrm{FPR}_{\mathrm{hold}}$ ($\sqrt{\cdot}$)
80%
50%
20%
5%
1%
0%
$N_{cal}=500$
100
500
2.5k
15k
Calibration Windows N_{cal} ($\log_{10}$)